

MOHD SAHIL ANSARI

DATA SCIENCE | MACHINE LEARNING | FRAUD & RISK ANALYTICS

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PROFESSIONAL SUMMARY

Computer Science graduate moving into Data Science and Machine Learning, with 2+ years of hands-on experience in fraud, chargeback and AML/KYC investigations. Builds machine learning projects in Python (Pandas, Scikit-learn, SHAP, Flask), including an explainable fraud detection and risk scoring system. Looking to apply financial-crime domain knowledge to Data Science, Machine Learning, Fraud Analytics and Risk Analytics roles.

TECHNICAL & DOMAIN SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Matplotlib, Exploratory Data Analysis

Machine Learning: Classification, Random Forest, Logistic Regression, Feature Engineering, Model Evaluation, Imbalanced Classification

NLP: Text Preprocessing, TF-IDF, CountVectorizer, Cosine Similarity

Tools & Explainability: Scikit-learn, SHAP, Flask, Git, GitHub

Financial Crime: Fraud Investigation, AML, KYC/CDD, PEP & Sanctions Screening, Transaction Monitoring, Visa Chargebacks, Dispute Resolution, Risk Assessment

PROFESSIONAL EXPERIENCE

Fraud & Chargeback Analyst (Customer Service Associate) | Teleperformance | Mumbai | Jul 2025 – Present

- Investigate card-fraud and transaction-dispute cases for a Germany-based bank client by reviewing transaction details, customer information and supporting evidence.
- Make evidence-based, risk-aware decisions on fraudulent and non-fraudulent cases while meeting quality and timeline standards.
- Initiate Visa chargebacks for eligible disputes after validating the evidence and case eligibility.

AML Analyst (Customer Service Associate) | Firstsource | Mumbai | Jul 2024 – Jul 2025

- Performed AML/KYC reviews for a UK-based client: customer due diligence, identity document verification, PEP and sanctions screening, and risk assessment.
- Handled high-volume name-screening reviews (approximately 200 cases daily) within strict SLA timelines.
- Reviewed real-time alerts and transaction/activity indicators to support account-opening and investigation decisions, escalating cases that needed further review.

SELECTED PROJECTS

Fraud Detection & Transaction Risk Scoring System | Python, Pandas, Scikit-learn, SHAP, Flask | PAYSim (synthetic)

- Built an end-to-end fraud detection and investigation workflow on the synthetic PAYSim financial transaction dataset.
- Engineered historical behavioural features and used chronological train/validation/test splits to reduce data leakage; handled class imbalance during training.
- Trained a Random Forest classifier and evaluated it with precision, recall, F1, PR-AUC, ROC-AUC and confusion matrix; generated transaction-level risk scores.
- Used SHAP TreeExplainer for model interpretability and added investigation guidance in a Flask interface.
- Built for demonstration on synthetic data; not tested on real banking data.

Movie Recommendation System | Python, Pandas, Scikit-learn, NLTK, Flask, TMDB API

- Built a content-based recommender on the TMDB 5000 dataset using CountVectorizer and cosine similarity, with text preprocessing and Porter stemming.
- Deployed it as a Flask web app with TMDB API integration for movie details and posters.

SMS Spam Classifier | Python, Pandas, Scikit-learn, NLTK

- Built an NLP pipeline with text cleaning, stemming and TF-IDF features to classify SMS messages as spam or not spam.
- Evaluated models with precision, recall, F1-score and confusion matrix.

EDUCATION & CERTIFICATION

B.Sc. Computer Science | University of Mumbai | Sep 2020 – Apr 2023 | Grade: A+

Data Science with Generative AI | Physics Wallah | Issued May 2025